Analysis of optical microwave properties and their use in interferometry

Timo Lubas and Julien Spitzlay

We examined the angular distribution of a microwave transmitter and measured the FWHM with $\gamma = (12.8 \pm 1.4)^\circ$. We also measured the wavelength with a standing wave $\lambda_{SW} = (3.256 \pm 0.022)\,\mathrm{cm}$ and a Michelson interferometer to $\lambda_{MI} = (3.286 \pm 0.061)\,\mathrm{cm}$. Next, we found out that the sender transmitted vertically polarized waves and that the waveguide used here does not change the polarization during the wave's propagation through its inside. Additionally, we checked Malus's law and measured a correspondence with extra attenuation. Furthermore, we found the expected exponential decay of an evanescent wave's intensity only for a small interval. We determined the lattice constant of a crystal model with two crystal planes to be $a_{100} = (4.00 \pm 0.32)\,\mathrm{cm}$ and $a_{110} = (4.04 \pm 0.21)\,\mathrm{cm}$.

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1 Introduction

The understanding of electromagnetic waves is an important aspect in physics, especially because of analysis methods in solid-state physics based on wave properties. In this experiment we use microwaves which have a wavelength of some centimeters so that we can work with macroscopic objects for examination.
Firstly, we measured the characteristics of the transmitter - a *Gunn diode* - namely the angular distribution and polarization. Furthermore, the optical properties of the microwaves were studied with a lens, a Michelson interferometer, a prism and a waveguide. Another examined optical phenomenon was the total internal reflection, according to quantum mechanics, also known as (quantum) tunneling. In the last part of the experiment we used some of those properties for Bragg reflection on a crystal model as visualization of the interaction of wave and solid.

2 Angular distribution of microwaves

To study the angular distribution the transmitter was installed on a goniometer and the receiver was rotated with a constant distance around the Gunn diode. The effective point of emission was positioned on the rotating axis. During the rotation the voltage was measured depending on the angle. This voltage is proportional to the absolute square of the electromagnetic wave, in other words to its intensity.
We used the transmitter *Valvo CL8632* which, according to the manufacturer's data, emits microwaves with $\nu_{MD} = 9.35\,\mathrm{GHz} \Leftrightarrow \lambda_{MD} = 3.21\,\mathrm{cm}$ ([2]). The used voltmeter was the *Agilent-34405a*.

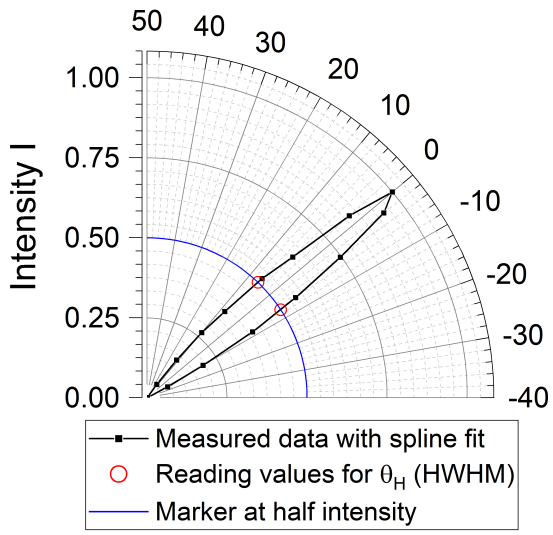

Fig. 1: This plot shows the intensity I (normalized to 1) and its angular distribution. Additionally, the level of half intensity was marked and the two values for the HWHM θ_H determined with an estimated angular uncertainty of 2°.

Fig.1 is a plot of the measurement (normalized to

1) in polar coordinates and shows a strict symmetry around 0°. To quantify the stretch of the radiation we computed the *full width at half maximum (FWHM)* γ, which is the double of the angle at the *half width at half maximum (HWHM)* θ_H. We assumed an estimated error of 1° and used Gaussian propagation of uncertainty which yields:

$$2\theta_H = \gamma = (12.8 \pm 1.4)^\circ$$

3 Methods for measuring the wavelength

3.1 Standing wave

Standing waves are the result of two interfering waves propagating in opposite directions. We used a lens to focus the radiation to a parallel beam and a metal plate to reflect the beam back on itself. Due to this composition we observed a standing wave between the lens and the plate, mathematically described by:

$$A(x,t) = 2A_0 \sin(2\pi\nu t)\cos(2\pi x/\lambda) \tag{1}$$

The nodes of this wave have a fixed position and the distance between two directly adjacent nodes is $\lambda/2$. This means that the wavelength can be computed by measuring a distance d and counting the maxima of intensity. However, to calculate the wavelength the number of antinodes n is relevant, which is the count of maxima decremented by one. This leads to:

$$\lambda = \frac{2d}{n} \tag{2}$$

On a distance of $d = (63.50 \pm 0.42)$ cm we counted 40 nodes which corresponds to 39 antinodes. Using eq. 2 yields to:

$$\lambda_{SW} = (3.256 \pm 0.022)\,\text{cm}$$

We estimated a reading error of 0.3 cm and used Gaussian propagation of uncertainty to calculate the errors of d and λ_{sw}. This value is not compatible with the manufacturer information.

3.2 Michelson interferometer

Another method for measuring a wavelength is a *Michelson interferometer* (fig. 2). This tool splits a beam up into two rays which travel orthogonal and are reflected by two metal plates. One of these is adjustable so that the distance to the beam splitter is variable with an *optical retardation* of $2d$, where d is the distance between the plate positions. After the reflection both beams travel back to the beam splitter and interfere constructively or deconstructively, depending on the retardation, which leads to maxima and minima.

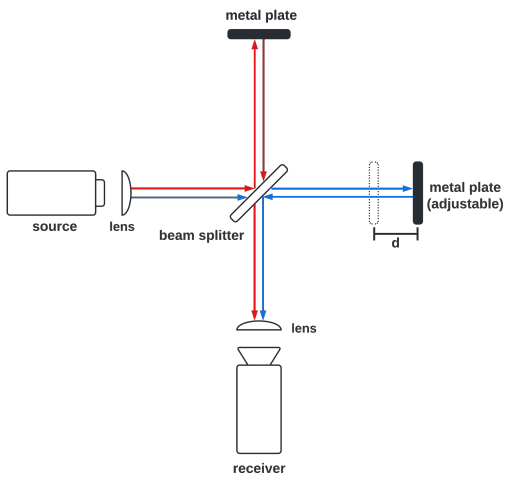

Fig. 2: This is the setup of a Michelson interferometer. The beam splitter is a hardboard and acts as semipermeable mirror. By changing the retardation d one can measure the interference with the receiver.

During the reduction of the distance $d = (23.00 \pm 0.42)$ cm we counted 14 maxima. Considering that the microwave has to travel the retardation two times, which leads again to eq. 2 we determined the wavelength to be:

$$\lambda_{MI} = (3.286 \pm 0.061)\,\text{cm}$$

We used the same reading error of 0.3 cm as before. This value is compatible with λ_{SW} but not with the manufacturer data λ_{MD}.

4 Focal length of a wax lens

Lenses are a useful tool in optics. An appropriate material to collimate microwaves is wax. We used the *thin lens equation*, known from geometrical optics, to compute the *focal length* f of a wax lens. With the definitions of the *object distance* g and the *image distance* b the lens equation has the following form ([4]):

$$\frac{1}{f} = \frac{1}{g} + \frac{1}{b} \tag{3}$$

For determining the image distance, we had to find the maximum of intensity behind the lens. Fig.3 shows the measured data. One can identify an additional overlying oscillation, which probably caused by standing waves, created by the wall behind the setup. We tried to compensate this problem by using the peak maxima for fitting a Lorentz function. The result is plotted in fig.4 and shows an intensity maximum at the position

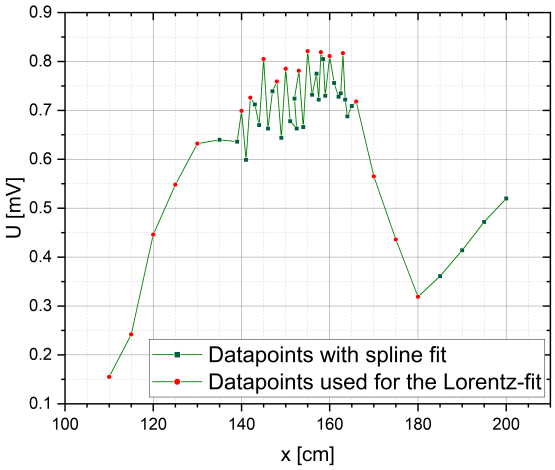

Fig. 3: This plot shows the measured intensity depending on the position of the detector on the optical bench. One can identify an overlying oscillation near the peak of the distribution. The red points mark the values chosen for the fit.

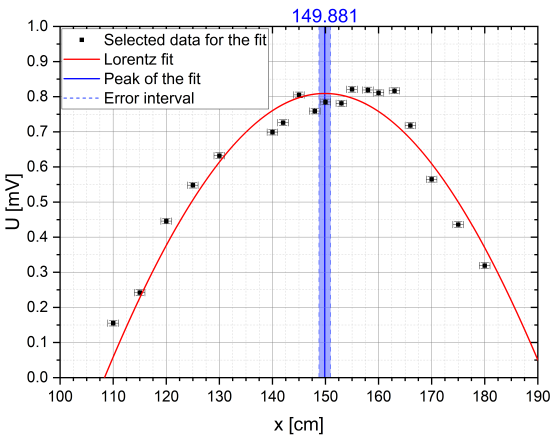

Fig. 4: It is shown a plot of the measuring points used, (mostly the peaks of the oscillation, cf. 3) with error bars and the weighted Lorentz fit. Its maximum describes the peak of intensity on the optical bench. The errors are calculated by estimated and instrumental errors.

$$x = (149.9 \pm 1.1)\,\mathrm{cm}$$

on the optical bench. This value and its uncertainty are regression parameters from the Lorentz fit. By measuring the devices' position and width we could compute object and image distance:

$$g = (77.7 \pm 1.4)\,\mathrm{cm}$$
$$b = (71.2 \pm 2.0)\,\mathrm{cm}$$

The errors were calculated with the estimated uncertainties of the positions of the transmitter, the lens and the receiver (all $\pm 1.0\,\mathrm{cm}$) and of their width (all $\pm 0.3\,\mathrm{cm}$). Furthermore, we used the manual of the multimeter ([3]) to calculate the errors of voltage for the weighted fit.
Eq.3 and Gaussian error propagation yield for the focal length f:

$$f = (37.13 \pm 0.63)\,\mathrm{cm}$$

It is important to mention that we used the thin lens equation for simplicity. This leads to an additional uncertainty. In all the other setups we tried to position the lens in the appropriate distance to the transmitter and receiver concerning the measured focal length.

5 Determination of the polarization

Electromagnetic waves, like every transversal wave, can be linearly, circularly or elliptically polarized. These states are manipulable with optical devices e.g. filters. In this section, we used a metal grid as polarizer. It is permeable for the component of the electric field which oscillates orthogonal to the wires of the grid.
The *intensity* I of a linearly polarized electromagnetic wave, depending on the angle θ between the initial polarization direction and the axis of the polarizer, is described by *Malus's law* ([4]):

$$I(\theta) = I_0 \cos^2\theta \tag{4}$$

It is important to know, that the receiver works as polarizer as well. Using Malus's law a second time leads to the detected intensity with the angle θ_1 between the initial polarization direction and the axis of the polarizer and θ_2 between the polarization direction behind the grid and the receiver axis.

$$I(\theta) = I_0 \cos^2(\theta_1)\cos^2(\theta_2) \tag{5}$$

We checked, that transmitter and receiver have to be orientated (anti-) parallel for measuring the maximal intensity. Additionally, we found that the source emits and the detector receives vertically polarized waves with reference to the table surface.
With eq.5 it is clear what happened when we installed a polarizer between transmitter and receiver and changed its orientation: If the sender and the receiver are perpendicular to another ($\theta_2 = 90° - \theta_1$) and a grid of vertical wires is positioned in the optical path, one detects no radiation. Rotating the grid leads to an intensity maximum for $\theta_1 = 45°$ and no signal for $\theta_1 = 90°$. If the transmitter and the receiver are orientated equally the situation simplifies because then the following applies: $\theta_1 = \theta_2$.

The last part of this experiment was a voltage measurement during a 180° rotation of the polarizer to verify Malus's law. Transmitter and receiver were orientated equally but we used the angle scale on the polarizer. This leads to a 90° rotation, because the polarizer's vertical orientation was marked with 0°. In other words, we expect a course that is proportional to $\sin^4(\beta)$ (cf. 5 with $90° - \theta_1 = 90° - \theta_2 = \beta$). Fig.5 compares the measured and theoretically expected course. The maximum was normalized to 1. It is recognizable that the measured data runs below the theoretical values. However, the shapes are similar and the maximum is at 90° in both cases. Probably, the polarizer is not a perfect filter and attenuates the intensity which explains the slimmer course.

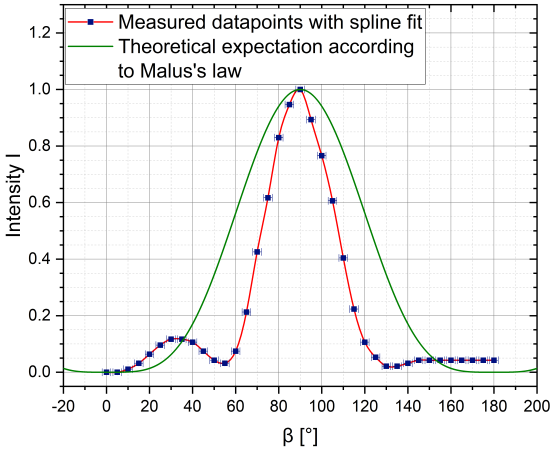

Fig. 5: This plot shows the measured data of voltage, proportional to the intensity, depending on the angle of a polarizer. In comparison to the course of Malus's law one can identify an extra attenuation because of the slimmer profile.

6 Microwaves in a waveguide

A waveguide is a resonator and uses reflections to direct the propagation of waves. In this part we tested the properties of a waveguide and its effect on microwaves. The sides of the waveguide - consisting of metal - force the direction of propagation because the electric field needs to have a node at the edges.
There are many categories to classify the accuring transversal modes. One important aspect is the orientation of the fields concerning the propagation direction. If the *E-field* is completely transversal and the *H-field* has a component parallel to the propagation direction, the mode is called *TE (transverse electric) mode*. The other possibility, called *TM (transverse magnetic) mode*, is that the H-field has no component in the direction of propagation, but this E-field component can have a value unequal to 0.

We installed the transmitter in front of the waveguide's mouth and the receiver in front of the other opening. With this structure we could measure a radiation at the end of the waveguide and by rotating the receiver also check that the waveguide preserved the polarization. By closing the waveguide with a cap, which eliminates radiation at the receiver, we proved that the waves propagate in the waveguide's interior.

7 Total internal reflection

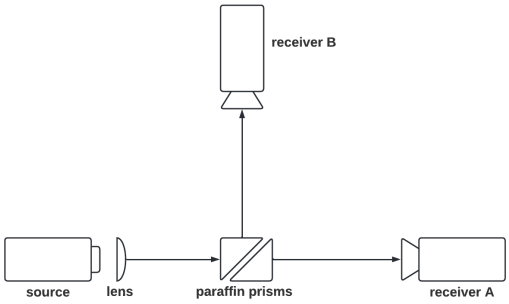

Fig. 6: This graphic shows the setup for measuring the transmitted (detected by receiver A) and reflected (detected by receiver B) ratios of a microwave after entering the paraffin prisms whose gap distance is adjustable.

Snell's law describes the angles θ_1 and θ_2 of incidence and refraction of electromagnetic waves travelling through two different isotropic media, therefore having two different refractive indexes n_1 and n_2 (([4]).

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \tag{6}$$

Assuming that $n_1 > n_2$, one can define a *critical angle* θ_C, which is the smallest angle of incidence that yields total reflection. This means $\theta_2 \geq 90°$, so in this case the wave does not leave the medium with higher refractive index. For calculating the critical angle one can use the equation:

$$\theta_C = \arcsin\frac{n_2}{n_1} \tag{7}$$

In geometrical optics the total reflection means a complete reflection of the wave. In reality however, the structure of waves leads to an exponential decay of an evanescent wave's intensity. The intensity I has the following dependency on the *optical penetration depth* s, in which z represents the distance from the surface ([1]):

$$I(z) \propto \exp\left(-\frac{2z}{s}\right) \tag{8}$$

with

$$s = \frac{\lambda_0 \cos\theta}{2\pi\sqrt{n_1^2 \sin^2\theta - n_2^2}} \tag{9}$$

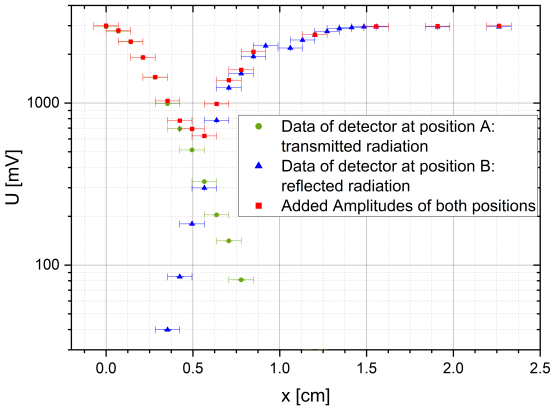

Fig. 7: This plot visualizes the measured voltage, proportional to the intensity of a transmitted (green) and reflected (blue) beam depending on the size of gap. The sum of both intensities is not constant and has a minimum. One can see an exponential decay in the transmitted beam in a short interval.

Fig.6 shows the experimental setup. The lens collimates the beam which interacts with the paraffin. Depending on the adjusted distance between the two prisms, the microwaves are transmitted (detected with receiver A) or reflected (detected with receiver B).
Fig.7 is a plot of the measured intensity on a logarithmic scale depending on the size of gap. In the interval from 0.2 cm to 0.6 cm one can identify a linear decay of the transmitted radiation. This implies a validation of the exponential decay (cf. 9). With a larger gap, the exponential decay describes the radiation insufficiently.
Furthermore, one can recognize that the addition of the two intensities is not constant. At the crossing point of the two measured voltages, the sum of both has its minimum. Reasons for these deviations from expectations could be multiple reflections between the gap and the fact, that no second lens in front of the detector was used.

8 Bragg reflection

In solid-state physics the lattice constant of a crystal is an important characteristic. Interference experiments are helpful to determine these short distances. Constructive interference accrues, if the optical retardation is a multiple of the half wavelength:

$$n\lambda = 2d_{hkl}\sin\theta \tag{10}$$

Here, d_{hkl} is the lattice plane distant. Its link to the lattice constant a is given by:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \tag{11}$$

Eq.11 only applies for cubic crystals. h, k and l are Miller indices which describe planes and directions in crystals.

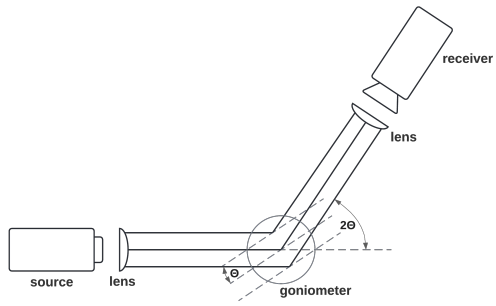

Fig. 8: Here, one can see the setup to determine the lattice constant of a crystal model, which is positioned on the goniometer. The lenses collimate the beam. By rotating the crystal and the receiver, one can measure the intensity of the Bragg reflection depending on the angle. Adapted from [1]

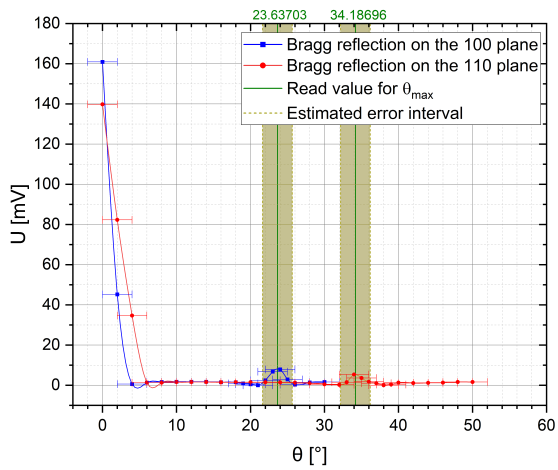

Fig. 9: This plot shows the measured the intensity, depending on the angle of incidence for the two planes (100) and (110) of a crystal. One can identify two peaks, which represent the Bragg reflection on that plane. With the read angles we calculated the lattice constant a.

Fig.8 shows the experimental setup. The crystal was orientated in such a way, that the examined plane was parallel to the optical path for $\theta = 0°$. We rotated the crystal with the angle θ, the receiver with 2θ and measured the voltage depending on the angle. The

results for the planes (100) and (110) are shown in fig.9. One can identify two peaks at the following positions:

$$\theta_{100} = (23.8 \pm 2.0)^\circ$$
$$\theta_{110} = (34.2 \pm 2.0)^\circ$$

The uncertainties are estimated based on the reading errors of the angle scale and the reading of the peaks by eye.
With eq.10 and 11 one can compute the lattice constant. Its error is a propagation of the angle uncertainty only, as we assumed an accurate manufacturer information concerning the wavelength.

$$a_{100} = (4.00 \pm 0.32)\,\mathrm{cm}$$
$$a_{110} = (4.04 \pm 0.21)\,\mathrm{cm}$$

Both results are compatible within their uncertainty interval.

9 Conclusion

In this experiment we measured the properties of a microwave transmitter and used the optical quantities of microwaves to gauge a crystal's lattice constant. The angular distribution has a strict symmetry around 0° and a FWHM of $(12.8 \pm 1.4)^\circ$. Next, we measured the transmitted wavelength with two different methods. First, we produced a standing wave and counted the nodes, which led us to a wavelength of $\lambda_{SW} = (3.256 \pm 0.022)\,\mathrm{cm}$. For the second method, a Michelson interferometer was used, that works with interference as well. We determined the wavelength to be $\lambda_{MI} = (3.286 \pm 0.061)\,\mathrm{cm}$.
Furthermore, we computed the focal length of a wax lens to be $f = (37.13 \pm 0.63)\,\mathrm{cm}$, by measuring the voltage. Especially in this part we experienced measuring difficulties, as other standing waves interfered with our signal.
Next, we found out that the sender transmits vertically polarized waves and the receiver works as polarizer, too. Additionally, we checked the properties of a waveguide. A microwave propagating through the inside of the waveguide does not change its polarization. The last experiment concerning polarization, was a comparison between our measurement of intensity depending on the angle of the polarizer and Malus's law. Except for an extra attenuation we found an accordance of theory and experiment.
An important wave phenomenon we examined was the total reflection and the evanescent wave. We found an exponential decay of the transmitted radiation, but could not see the constancy of the transmitted and reflected intensity accumulated.
Lastly, we studied Bragg reflection on a crystal model to determine its lattice constant. We measured these for the (100) and the (110) plane with the following values: $a_{100} = (4.00 \pm 0.32)\,\mathrm{cm}$ and $a_{110} = (4.04 \pm 0.21)\,\mathrm{cm}$.

References

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